# Mortgage Rates and Delinquency: Evidence from a Fuzzy Regression Discontinuity Design

Kirill Buryak[*] Daniil Kozhin[†] Margarita Kudaeva[‡]
Alexandr Shestakov[§] Yaroslav Volkov[¶]

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**Abstract**

This paper studies the effect of mortgage interest rates on loan delinquency using a fuzzy regression discontinuity design around the FICO 620 threshold. We exploit a discontinuity in mortgage pricing at the cutoff and compare borrowers close to the threshold. The validity checks indicate manipulation and covariate imbalance around the cutoff, suggesting that borrowers immediately above and below the threshold are not fully comparable. Across specifications, we do not find robust evidence that higher mortgage interest rates causally increase delinquency. Instead, the positive relationship between mortgage rates and delinquency appears to be driven primarily by borrower selection and underlying credit risk. The results also suggest that securitization may have affected mortgage markets through incentives for sorting and strategic behavior around institutional thresholds.

We did not use AI in preparation of this assignment.

[*]The New Economic School. Email: kburyak@nes.ru
[†]The New Economic School. Email: dkozhin@nes.ru
[‡]The New Economic School. Email: mkudaeva@nes.ru
[§]The New Economic School. Email: ashestakov@nes.ru
[¶]The New Economic School. Email: yavolkov@nes.ru

# 1 Data

Our dataset contains individual-level mortgage loan data, including borrower and loan characteristics. The borrower's credit score (FICO) serves as the running variable in the regression discontinuity design. The main outcome variable is delinquency, equal to one if the borrower is at least 60 days behind on payments. Observations with missing values for area are excluded because area is used for clustering standard errors. Logarithms of income and loan balance are used throughout the analysis. Table 1 reports summary statistics for the main variables; the average delinquency rate is 28% and the average mortgage interest rate is approximately 7%.

TABLE 1: Summary Statistics of Main Variables

| | Mean | Median | SD | Min | Max | N obs |
|---|---|---|---|---|---|---|
| **Panel A: Outcome Variable** | | | | | | |
| Delinquency (%) | 27.886 | 0.000 | 44.844 | 0 | 100 | 171299 |
| **Panel B: Loan Characteristics** | | | | | | |
| Interest Rate (%) | 6.933 | 7.000 | 2.824 | 0.010 | 18.875 | 171299 |
| Log Loan Balance | 12.070 | 12.196 | 0.792 | 9.210 | 15.420 | 171299 |
| Loan to Value | 67.925 | 78.575 | 22.964 | 1.335 | 102.999 | 171288 |
| Hard | 0.089 | 0.000 | 0.285 | 0.000 | 1.000 | 171299 |
| Refinance | 0.675 | 1.000 | 0.468 | 0.000 | 1.000 | 171299 |
| Broker | 0.927 | 1.000 | 0.259 | 0.000 | 1.000 | 171299 |
| **Panel C: Borrower Characteristics** | | | | | | |
| FICO | 641.747 | 646.000 | 21.921 | 570.000 | 670.000 | 171299 |
| Age | 43.537 | 43.000 | 12.233 | 18.000 | 99.000 | 161513 |
| Log Income | 8.731 | 8.723 | 0.664 | -4.605 | 15.341 | 138815 |
| Female | 0.621 | 1.000 | 0.485 | 0.000 | 1.000 | 163900 |
| White | 0.641 | 1.000 | 0.480 | 0.000 | 1.000 | 171299 |

Note: This table reports summary statistics for the main variables. Observations with missing values for area were removed from the sample prior to the analysis.

# 2 Empirical Strategy

The main empirical challenge in estimating the effect of mortgage interest rates on delinquency is the endogeneity of mortgage pricing. Lenders may use unobserved borrower characteristics when setting mortgage rates, implying that mortgage rates are correlated with default risk.

To address this issue, we exploit a discontinuity in mortgage pricing at the FICO score threshold of 620 and implement a fuzzy regression discontinuity design. Borrowers with FICO scores below the threshold tend to face higher mortgage interest rates. We estimate local fuzzy RDD specifications of the form:

$$Del_i = \alpha + \beta Rate_i + f(FICO_i - 620) + \theta X_i + \varepsilon_i$$

where $Del_i$ is an indicator for loan delinquency of borrower $i$, measured in percentage points, $Rate_i$ is the mortgage interest rate, $FICO_i$ is the borrower credit score, and $X_i$ includes borrower and loan characteristics.

Following Calonico et al. (2014), estimation is implemented using local linear regressions with a triangular kernel and mean squared error optimal bandwidth selection. As a sensitivity check, we additionally report estimates using a rectangular kernel. Standard errors are clustered at the area level.

The design is fuzzy because crossing the threshold changes mortgage pricing discontinuously but does not deterministically assign borrowers to a specific interest rate. Identification therefore relies on local comparability of borrowers around the cutoff, such that any discontinuous change in delinquency at the cutoff operates through mortgage pricing. Under this assumption, the fuzzy RDD estimates identify local average treatment effects for borrowers whose mortgage rates are affected by the threshold.

As an additional exercise, we estimate IV specifications:

$$Del_i = \alpha + \beta \widehat{Rate}_i + \delta_1(FICO_i - 620) + \delta_2(FICO_i - 620)^2 + \theta X_i + \varepsilon_i,$$

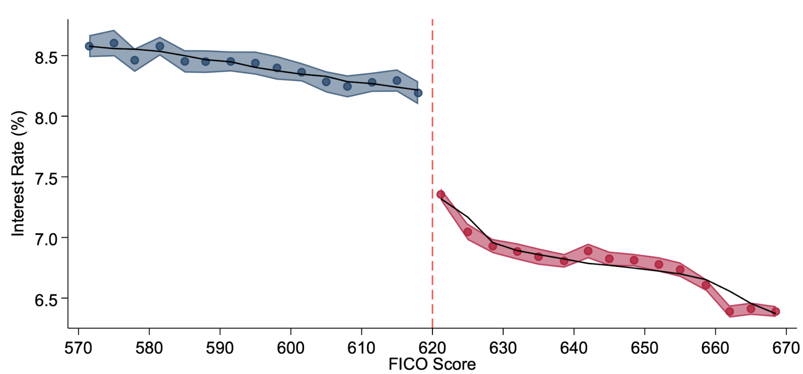
where $Rate_i$ is instrumented with an indicator for $FICO_i < 620$. Unlike the local RDD estimates, these specifications use observations farther from the cutoff and therefore should not be interpreted causally. However, they complement the local RDD estimates by assessing whether the relationship between mortgage rates and delinquency extends beyond borrowers near the cutoff and by indicating the likely direction and magnitude of endogeneity bias.

# 3 Identification and Validity Checks

## 3.1 Discontinuity in Mortgage Rates

A necessary condition for the fuzzy RDD design is a discontinuity in mortgage pricing at the FICO 620 threshold. Figure 1 shows a clear negative jump in mortgage interest rates at the cutoff. The corresponding first-stage estimates in Table 3 are economically

Figure 1: Interest Rate Discontinuity at the FICO Score 620 Threshold

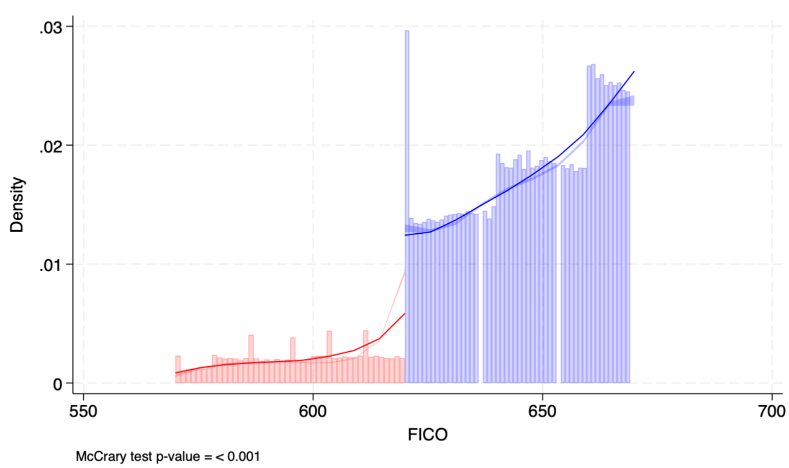



Note: This figure plots average interest rates by FICO score bins around the 620 threshold, indicated by the vertical dashed line. To improve visual clarity, the original bins are aggregated into 15 bins on each side of the cutoff. The shaded areas represent 95% confidence intervals for the bin means. The solid lines show fitted values from separate third-order polynomial regressions estimated on either side of the threshold.

large and statistically significant at the 1% level, confirming that the threshold strongly predicts mortgage pricing.

## 3.2 Density Tests

Figure 2: McCrary Density Test around the FICO 620 Threshold




Note: This figure presents the McCrary density test (McCrary, 2008) for manipulation of the running variable around the FICO score cutoff of 620. The figure plots estimated densities of the FICO score on either side of the threshold together with local polynomial fits.

Continuity of the running variable density around the cutoff is another important validity check. Following McCrary (2008), Figure 2 reports a statistically significant discontinuity in the density of FICO scores at the 620 threshold ($p < 0.001$). The bunching immediately above the cutoff suggests sorting or manipulation around the threshold.

One possible explanation is that borrowers manipulate FICO scores to qualify for better mortgage conditions. Another possibility is selective sample composition around the threshold, where borrowers below the cutoff are less likely to apply for loans or appear in the sample. However, as discussed by Jiang et al. (2014), manipulation around institutional thresholds does not necessarily invalidate identification if borrowers cannot precisely control the resulting credit score. Because credit score formulas are proprietary and outcomes of credit repair efforts are inherently noisy, strategic behavior may generate bunching near the cutoff without mechanically producing discontinuities in delinquency exactly at the threshold. Nevertheless, the significant density discontinuity suggests that observations around the cutoff may not be fully comparable, weakening the identifying assumptions of the RDD design.

## 3.3 Covariate Continuity

Another important validity check is continuity of observable predetermined borrower and loan characteristics around the cutoff. However, the results in Table 2 indicate statistically significant discontinuities in several covariates at the FICO 620 threshold.

TABLE 2: Covariate Balance around FICO 620 Cutoff

| | Log Loan Balance | Log Income | LTV | Broker | Hard | Refinance | Age | Female | White |
|---|---|---|---|---|---|---|---|---|---|
| RD Estimate | 0.204** | 0.137** | 1.757 | 0.052*** | 0.056*** | -0.180*** | -1.902*** | 0.063** | -0.053 |
| | (0.100) | (0.060) | (1.803) | (0.012) | (0.006) | (0.057) | (0.650) | (0.025) | (0.046) |
| Mean Dep Var | 12.052 | 8.669 | 69.943 | 0.927 | 0.081 | 0.616 | 42.934 | 0.623 | 0.631 |
| MSE-opt. bandwidth | 24.542 | 14.518 | 10.676 | 14.690 | 19.394 | 7.726 | 13.523 | 8.489 | 15.845 |
| Eff. Obs. | 73,484 | 34,769 | 31,518 | 42,610 | 56,611 | 23,301 | 37,675 | 25,022 | 45,358 |
| Left of cutoff | 10,487 | 6,125 | 4,454 | 6,197 | 8,490 | 3,089 | 5,593 | 3,433 | 6,641 |
| Right of cutoff | 62,997 | 28,644 | 27,064 | 36,413 | 48,121 | 20,212 | 32,082 | 21,589 | 38,717 |

Note: This table reports reduced-form regression discontinuity estimates of discontinuities in predetermined borrower and loan characteristics at the FICO 620 threshold. Each column uses the corresponding covariate as the dependent variable and FICO score as the running variable. Estimation uses the mean squared error optimal bandwidth reported in the table. The mean of the dependent variable is computed within the selected bandwidth. Standard errors clustered at the area level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

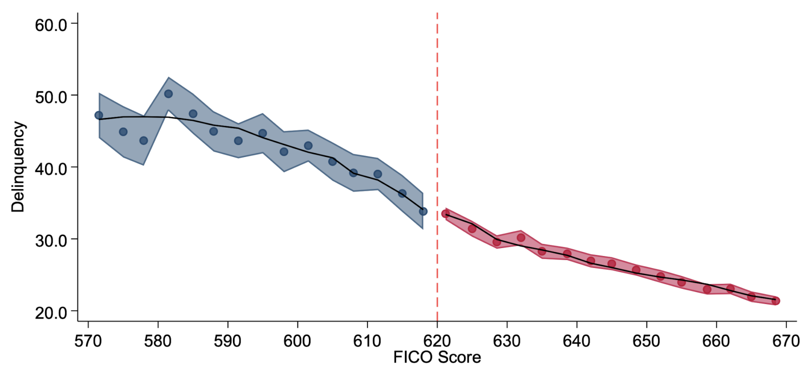
The largest discontinuities are observed for loan balance, income, the hard prepayment penalty indicator, and broker origination status. The graphical evidence in Figure A1 shows similar patterns. These results suggest that borrowers just above and below the cutoff differ systematically in observable characteristics. As a result, discontinuities in delinquency may partly reflect differences in borrower composition and loan characteristics rather than the causal effect of mortgage rates.

Overall, the validity checks provide substantial evidence against the standard identifying assumptions of the RDD design, suggesting that observations around the cutoff are not fully comparable and that the resulting estimates should be interpreted cautiously.

# 4 Main Results

Table 3 reports the main fuzzy RDD estimates of the effect of mortgage interest rates on delinquency. Across specifications, the estimated effects are statistically insignificant under both triangular and rectangular kernels. Although the first stage is strong, there is no visible discontinuity in delinquency at the cutoff (Figure 3). In addition, the estimates vary across specifications after including controls, suggesting sensitivity to differences in borrower and loan characteristics.

Figure 3: Delinquency Discontinuity at the FICO Score 620 Threshold




Note: This figure plots average delinquency rates by FICO score bins around the 620 threshold, indicated by the vertical dashed line. To improve visual clarity, the original bins are aggregated into 15 bins on each side of the cutoff. The shaded areas represent 95% confidence intervals for the bin means. The solid lines show fitted values from separate third-order polynomial regressions estimated on either side of the threshold.

These findings should be interpreted together with the validity concerns discussed above. The density and covariate tests indicate that borrowers immediately above and below the threshold are not fully comparable, implying that the baseline fuzzy RDD estimates may still reflect differences in borrower composition rather than only the causal effect of mortgage rates.

To address manipulation around the cutoff more directly, we additionally estimate donut RDD specifications that exclude observations immediately around the FICO 620 threshold. Following Barreca et al. (2011) and Noack and Rothe (2023), this approach reduces the influence of sorting in the running variable. The donut RDD estimates remain statistically insignificant, but become substantially more stable across specifications, converging to coefficients of approximately 1–1.5 percentage points. This suggests that the absence of a detectable local treatment effect is not driven solely by manipulation very close to the cutoff.

In contrast, the broader IV specifications reported in Table 4 produce large and statistically significant positive estimates, reaching approximately 5 percentage points in the

TABLE 3: Fuzzy RDD Estimates of the Effect of Mortgage Interest Rates on Delinquency

| | Delinquency (%) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Triangular Kernel | | | Rectangular Kernel | | | Donut RDD | | |
| Interest Rate (%) | -3.028<br>(4.091) | 0.505<br>(3.885) | 0.884<br>(3.872) | -2.815<br>(4.261) | 0.259<br>(3.019) | 1.325<br>(3.100) | 1.276<br>(1.813) | 1.553<br>(2.729) | 1.115<br>(2.856) |
| First-stage jump in rate | -0.584*** | -0.567*** | -0.539*** | -0.584*** | -0.620*** | -0.582*** | -1.179*** | -0.885*** | -0.825*** |
| First-stage SE | (0.192) | (0.119) | (0.108) | (0.192) | (0.113) | (0.102) | (0.168) | (0.107) | (0.094) |
| Loan Controls | No | Yes | Yes | No | Yes | Yes | No | Yes | Yes |
| Borrower Controls | No | No | Yes | No | No | Yes | No | No | Yes |
| MSE-opt. Bandwidth | 12.723 | 12.199 | 12.859 | 12.723 | 11.003 | 12.408 | 12.162 | 11.141 | 11.431 |
| Eff. Obs. | 37,081 | 30,228 | 28,568 | 37,081 | 27,968 | 28,568 | 29,803 | 22,051 | 20,800 |
| Left of cutoff | 5,328 | 5,262 | 5,057 | 5,328 | 4,818 | 5,057 | 4,908 | 4,400 | 4,222 |
| Right of cutoff | 31,753 | 24,966 | 23,511 | 31,753 | 23,150 | 23,511 | 24,895 | 17,651 | 16,578 |

Note: This table reports fuzzy regression discontinuity estimates of the effect of mortgage interest rates on loan delinquency, where delinquency is measured in percentage points. The running variable is the borrower FICO score, with a cutoff at 620, and the treatment variable is the mortgage interest rate. Columns 1–3 use a triangular kernel, columns 4–6 use a rectangular kernel, and columns 7–9 report donut RDD estimates after excluding observations with FICO scores equal to 619, 620, and 621. Estimation is based on local linear regressions using the mean squared error optimal bandwidth reported in the table. The first-stage estimates report the discontinuity in mortgage interest rates at the cutoff. Standard errors clustered at the area level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

most demanding specification. However, unlike the local RDD design, these estimates compare borrowers with systematically different risk profiles. Borrowers with lower FICO scores tend to receive both higher mortgage rates and exhibit higher baseline delinquency risk. Taken together, the insignificant local RDD estimates and large positive IV estimates suggest that the positive relationship between mortgage rates and delinquency is likely driven, at least in part, by borrower selection and unobserved risk characteristics rather than by a causal effect of mortgage rates alone.

More broadly, the results suggest that securitization may have affected mortgage markets through channels beyond the screening mechanism emphasized by Keys et al. (2010). The significant density discontinuity and covariate imbalance around the FICO 620 threshold indicate sorting and strategic behavior around institutional cutoffs. In particular, the securitization rules tied to the FICO threshold may have created incentives for borrowers and lenders to manipulate or strategically target scores around 620 in order to obtain better mortgage conditions and easier access to securitization. Similar concerns are discussed by Jiang et al. (2014), who argue that discontinuities around securitization thresholds may partly reflect endogenous borrower composition and strategic behavior in credit markets. At the same time, despite a strong discontinuity in mortgage pricing, the local causal effect of mortgage rates on delinquency remains weak and statistically insignificant. Taken together, these findings suggest that securitization may have distorted incentives, borrower composition, and pricing around institutional thresholds rather than increasing delinquency primarily through mortgage rates themselves.

TABLE 4: OLS and IV Estimates of the Effect of Mortgage Interest Rates on Delinquency

| | Delinquency (%) | | | | | |
|---|---|---|---|---|---|---|
| | OLS | | | IV | | |
| Interest Rate (%) | 2.045*** | 2.056*** | 2.006*** | 2.846** | 4.773*** | 5.250*** |
| | (0.104) | (0.187) | (0.182) | (1.218) | (1.659) | (1.679) |
| Loan Controls | No | Yes | Yes | No | Yes | Yes |
| Borrower Controls | No | No | Yes | No | No | Yes |
| Mean Dep Var | 28.35 | 28.35 | 28.35 | 28.35 | 28.35 | 28.35 |
| R-squared | 0.017 | 0.033 | 0.036 | 0.028 | 0.028 | 0.023 |
| Number of Clusters | 901 | 901 | 901 | 901 | 901 | 901 |
| First-stage F-stat | | | | 40.84 | 44.06 | 45.67 |
| Observations | 130,265 | 130,265 | 130,265 | 130,265 | 130,265 | 130,265 |

Note: This table reports estimates of the effect of mortgage interest rates on loan delinquency, where delinquency is measured in percentage points. Columns 1–3 present ordinary least squares (OLS) estimates, while columns 4–6 report instrumental variables estimates from a fuzzy regression discontinuity design in which the mortgage interest rate is instrumented using an indicator for borrowers with FICO scores below the 620 threshold. The IV specifications control for a quadratic polynomial in the centered FICO score. Coefficients therefore represent percentage-point changes in delinquency associated with a one percentage-point increase in the mortgage interest rate. Standard errors clustered at the area level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

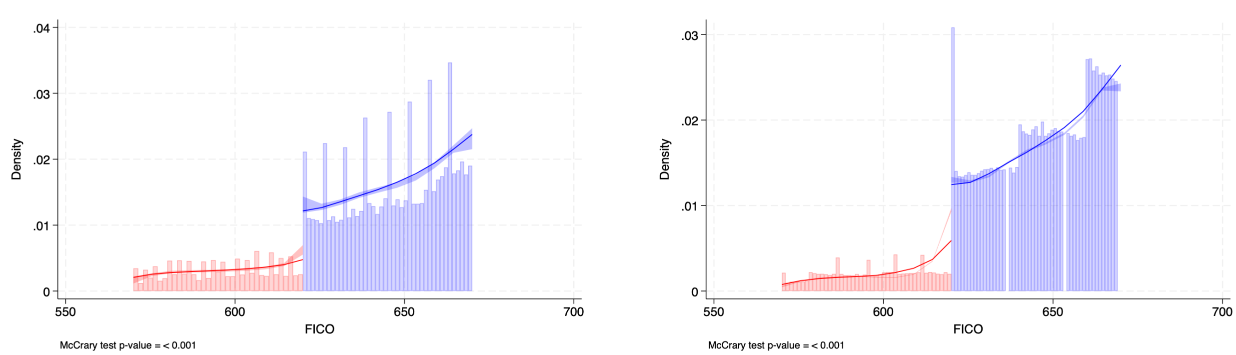
## 5 Heterogeneity by Origination Channel

We next examine whether manipulation around the cutoff differs across loan origination channels. Figure 4 shows substantially stronger bunching around the FICO 620 threshold for broker-originated loans relative to bank-originated loans. This pattern suggests stronger incentives within the broker channel to strategically target the securitization threshold.

Figure 4: McCrary Density Tests by Loan Origination Channel

Panel A: Bank-originated loans

Panel B: Broker-originated loans




Note: This figure presents McCrary density tests around the FICO score cutoff of 620 separately for bank-originated and broker-originated loans. Each panel plots estimated densities of the running variable on either side of the threshold together with local polynomial fits. The vertical dashed line indicates the cutoff.

At the same time, the first-stage estimates reported in Table 5 are larger for bank-originated loans, suggesting that pricing rules tied to the threshold were applied more rigidly within the banking channel. Despite these differences, the estimated effects of mortgage rates on delinquency remain statistically insignificant across specifications for both subsamples.

TABLE 5: Fuzzy RDD Estimates of the Effect of Mortgage Interest Rates on Delinquency

| | Delinquency (%) | | | | | |
|---|---|---|---|---|---|---|
| | Bank | | | Broker | | |
| Interest Rate (%) | 2.231<br>(4.186) | 1.273<br>(4.832) | 1.051<br>(4.961) | -2.493<br>(3.990) | 0.522<br>(3.966) | 1.043<br>(4.100) |
| First-stage jump in rate | -1.108*** | -0.835*** | -0.823*** | -0.580*** | -0.571*** | -0.537*** |
| First-stage SE | (0.215) | (0.131) | (0.127) | (0.195) | (0.124) | (0.113) |
| Loan Controls | No | Yes | Yes | No | Yes | Yes |
| Borrower Controls | No | No | Yes | No | No | Yes |
| MSE-opt. Bandwidth | 12.959 | 15.557 | 15.529 | 12.943 | 12.406 | 12.644 |
| Eff. Obs. | 2,689 | 2,839 | 2,771 | 34,392 | 28,009 | 26,396 |
| Left of cutoff | 600 | 764 | 747 | 4,728 | 4,672 | 4,479 |
| Right of cutoff | 2,089 | 2,075 | 2,024 | 29,664 | 23,337 | 21,917 |

Note: This table reports fuzzy regression discontinuity estimates of the effect of mortgage interest rates on loan delinquency separately for bank-originated and broker-originated loans. Delinquency is measured in percentage points. The running variable is the borrower FICO score, with a cutoff at 620, and the treatment variable is the mortgage interest rate. Estimation is based on local linear regressions using the mean squared error optimal bandwidth reported in the table. The first-stage estimates report the discontinuity in mortgage interest rates at the cutoff. Standard errors clustered at the area level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

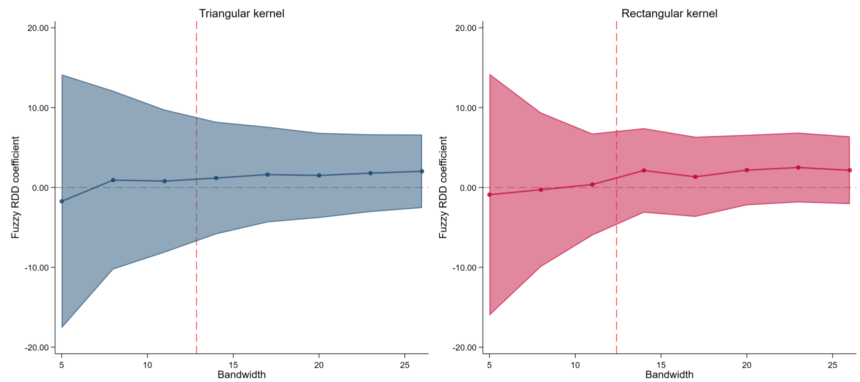
Overall, the heterogeneity results suggest that strategic behavior around securitization thresholds was concentrated more strongly among broker-originated loans. At the same time, the absence of statistically significant local treatment effects in both subsamples indicates that the main results are not driven by a particular origination channel.

## 6 Sensitivity Analysis

Finally, we examine whether the main results are sensitive to alternative bandwidth choices. Figure 5 shows that the main results are not sensitive to bandwidth choice. For both triangular and rectangular kernels, the estimated effect of mortgage interest rates on delinquency remains statistically insignificant across all considered bandwidths. Although precision increases at larger bandwidths, the confidence intervals continue to include zero.

Overall, the sensitivity analysis supports the baseline findings and does not provide robust evidence that higher mortgage rates increase delinquency around the FICO 620 threshold.

Figure 5: Sensitivity of Fuzzy RDD Estimates to Bandwidth Choice




Note: This figure reports fuzzy RDD estimates of the effect of mortgage interest rates on delinquency across alternative bandwidth choices for triangular and rectangular kernels. The solid lines represent point estimates and the shaded areas denote 95% confidence intervals. The dashed vertical lines indicate the mean squared error optimal bandwidths: 12.86 for the triangular kernel and 12.41 for the rectangular kernel.

## 7 Conclusion

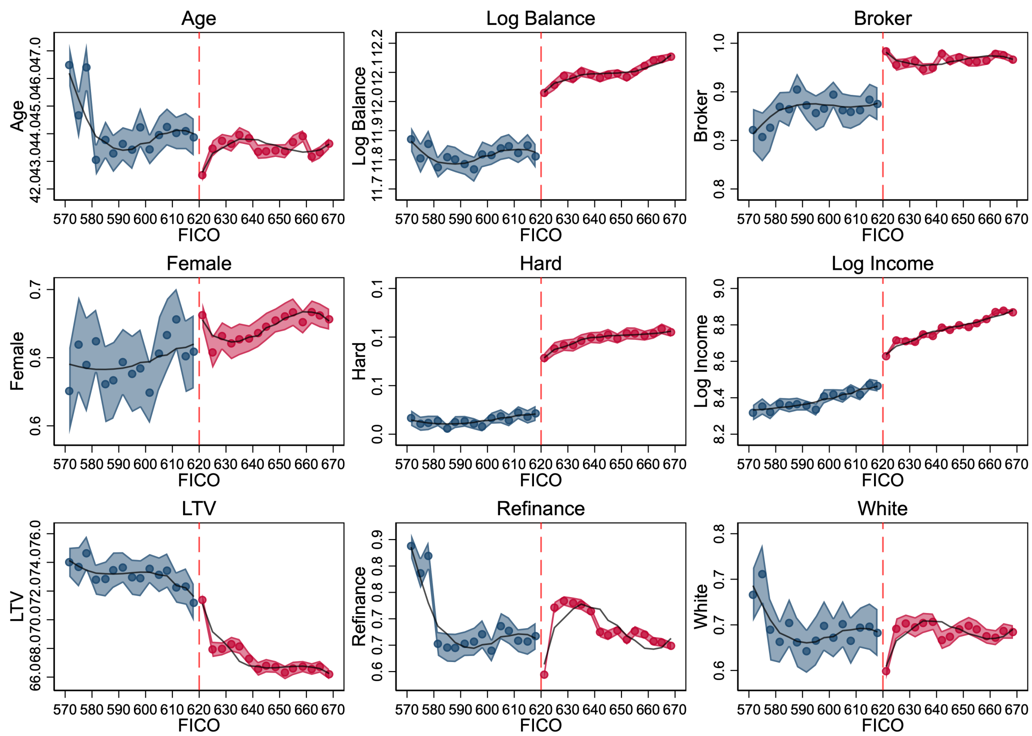
This paper studies the effect of mortgage interest rates on delinquency using a fuzzy regression discontinuity design around the FICO 620 threshold. Although the first stage is strong, the validity checks indicate sorting and covariate discontinuities around the cutoff, suggesting that observations near the threshold are not fully comparable.

The local fuzzy RDD estimates do not provide robust evidence that higher mortgage interest rates increase delinquency. In contrast, IV estimates are large and positive, suggesting that the observed relationship between mortgage rates and delinquency is driven primarily by borrower selection and underlying credit risk.

More broadly, the results suggest that securitization may have created incentives for sorting and strategic behavior around institutional thresholds rather than increasing delinquency directly through mortgage pricing.

# 8 Appendix

Figure A1: Covariate Balance Plots Around the FICO 620 Threshold




Note: This figure presents average borrower and loan characteristics by FICO score bins around the 620 threshold. To improve visual clarity, the original bins are aggregated into 15 bins on each side of the cutoff in each panel. The shaded areas represent 95% confidence intervals for the bin means, while the solid lines show fitted values from separate third-order polynomial regressions estimated on either side of the threshold. The vertical dashed line indicates the cutoff.

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